

cswhite: Chen & Storey whitening with KING kinship (assumes HWE)

genotypes G

KING kinship $\geq \tau$

Σ (families)

$\Sigma^{-1/2}GG'\Sigma^{-1/2}$

PCs (whitened)

KING-robust

families,
 $v = D$

whiten

eigvecs,
 $\times \Sigma^{1/2}$

PC3

PC2

● CEU ● CHB ● MXL ● YRI ★ relative

